

Implementing the 2021 UNESCO Recommendation on Open Science

Qatar's National Report

Introduction

This report outlines Qatar's progress, challenges, and future directions in implementing the 2021 UNESCO Recommendation on Open Science. It was prepared collaboratively by key national stakeholders, including Qatar University (QU), the Ministry of Education and Higher Education (MEHE), The Qatar Research, Development, and Innovation Council (QRDI Council), and the University of Doha for Science & Technology. The report reflects a comprehensive consultation process undertaken in response to UNESCO's mandate. Guided by the seven areas of action outlined in the UNESCO Recommendation and the associated Questionnaire for Implementation, this consultation involved mapping existing policies, mechanisms, and activities related to the core pillars of open science: scientific knowledge, infrastructure, communication, engagement of societal actors, and open dialogue with other knowledge systems. The report aims to: (i) assess Qatar's current landscape against the objectives of the Recommendation; (ii) document progress and highlight best practices in open science; and (iii) identify key challenges and opportunities for further implementation, including specific capacity-building needs. Ultimately, this report contributes to UNESCO's broader effort to promote the open availability, accessibility, and reusability of multilingual scientific knowledge across all disciplines and scholarly practices, ensuring that the benefits of science are shared by all.

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1. Promotion and Sharing of the 2021 Recommendation with National Ministries and Institutions

The ministries and national authorities and entities involved in the promotion of the 2021 Recommendation include Qatar University (QU), the Ministry of Education and Higher Education (MEHE), Qatar National Research Fund (QNRF), the QRDI Council, Qatar Foundation (QF), and University of Doha for science & Technology.

To illustrate the promotion and sharing of the 2021 Recommendation, we highlight the efforts of Qatar University (QU) and the Qatar Research, Development, and Innovation Council (QRDI Council).

Qatar University:

While the 2021 UNESCO Recommendation on Open Science has not been directly disseminated or promoted within Qatar University, the institution's general context and involvement in open science demonstrate noteworthy efforts and approaches, as follows:

Institutional Policies and Guidelines: Qatar University has established various academic and research guidelines. The principles of the 2021 Recommendation are being indirectly integrated into these guidelines to promote open access publishing and data sharing.

Qatar University Press: The QU Press Open Journal System for academic journals can be a crucial platform for advancing open access to research outputs from Qatar University.

Qatar University Library Initiatives: Qatar University Library is dedicated to promoting open science by providing access to Open Educational Resources (OER). This initiative allows faculty and students to access free and openly licensed educational materials. Additionally, the library manages the Qatar University institutional repository, which supports researchers in making their work publicly accessible. This repository serves as a vital infrastructure for open access by preserving and disseminating the university's scholarly output. The library plays a crucial role in raising awareness of open science principles and practices through various activities, including:

- **Workshops and Training:** Conducting sessions focused on open access, copyright, and related topics.
- **International Open Access Week Activities:** Participating in or promoting events during International Open Access Week to highlight the benefits of open access.
- **Information Dissemination:** Providing information through the library's website and other communication channels to educate the university community about open science.
- **Collaboration and Partnerships:** Working with other entities, such as university research centers and potentially international partners, to further advance open science initiatives.

The office for Digital Learning and Online Education: The Digital Learning and Online Education office at Qatar University is responsible for executing ambitions and strategic directions for digital learning and online education. This reflects a commitment to open access principles for educational materials.

Research and Graduate Studies Office: This office is responsible for establishing and overseeing research policies at QU. They are likely incorporating the values and principles of open science—such as open access to publications and the sharing of research data—into updated guidelines for researchers and graduate students. This may be reflected in requirements or recommendations concerning research plans, publication venues, and intellectual property considerations.

The Qatar Research, Development, and Innovation Council

The Qatar Research, Development, and Innovation Council (QRDI Council) is a national organization responsible for leading and promoting scientific research, development, and innovation in Qatar. In line with Qatar's National Vision 2030 and associated strategies, the Council aims to foster a knowledge-based economy by facilitating collaboration among stakeholders, promoting advanced research, and enhancing innovation ecosystems. Committed to advancing open science, the QRDI Council collaborates with ministries, universities, research institutes, and industry partners to integrate open science principles across various sectors. The Council focuses on several key areas, specifically:

Funding and Enabling RDI Activities: The QRDI Council provides funding for research, development, and innovation projects involving universities, ministries, and the private sector. This support encourages collaboration, promotes excellence, and aligns initiatives with national priorities, especially those that adhere to open science principles.

Raising Awareness and Building Consensus: Through strategic dialogues, workshops, and national consultations, the QRDI Council collaborates with stakeholders to foster a shared understanding of open science principles, which encompass accessibility, reproducibility, collaboration, and ethical data use.

Highlighting the Benefits: The Council effectively communicates the diverse benefits of open science, including increased research visibility, accelerated innovation, enhanced international collaboration, and improved returns on public investment in R&D.

Addressing Challenges: QRDI Council recognizes and aims to tackle the challenges of open science, such as data privacy issues, intellectual property rights, and the necessity for infrastructure, capacity building, and suitable policy frameworks.

Encouraging Diverse Pathways to Open Science: Recognizing that open science is not a one-size-fits-all model, the Council supports diverse approaches tailored to different disciplines, institutions, and sectors. This includes promoting open access publishing, open data repositories, citizen science initiatives, and collaborative platforms that foster transparency and knowledge sharing.

Contributing to Knowledge Dissemination: QRDI Council contributes to national and international knowledge dissemination efforts by supporting the publication of research outputs in open formats, facilitating data sharing agreements, and encouraging practices that democratize access to scientific knowledge. One such event is the Research Outcome Seminars (ROSs) where the council periodically ask its grant holders disseminate their findings to the public. Such event creates a real opportunity to engage public in scientific debates.

Regular updates of Policies: As part of its commitment to transparency and alignment with international best practices, the QRDI Council undertook a significant policy update in 2022. The Council revised its funding policies for both scientific research and talent development tracks, introducing new requirements that are consistent with Open Science Guidelines. These updated policies place a strong emphasis on transparency, the open dissemination of knowledge, and the promotion of open-access outputs and data sharing. By fully embedding open science principles into its funding framework, the Council aims to ensure that publicly funded research delivers maximum impact, is accessible to the broader scientific community, and supports knowledge transfer within and beyond Qatar. To reinforce this commitment, the Council has made these policies publicly available in both English and Arabic, ensuring accessibility for all stakeholders and enhancing accountability in the national RDI system.

Online resources-Innolight: The Council has also developed a dedicated online platform that serves as a central hub for information and resources related to its programs and policies. This platform provides comprehensive access to key documents, including funding guidelines, strategic initiatives, and updates on national research and innovation activities. In support of open science and knowledge dissemination, the platform features a repository of research publications produced by researchers in Qatar, helping to increase the visibility and accessibility of local scientific outputs. Additionally, it includes a catalogue of available research infrastructure/ equipment across institutions in the country, enabling researchers to identify and access shared resources and facilities that can support their work. This initiative is designed to foster collaboration, reduce duplication, and maximize the use of existing national RDI assets. In doing so, QRDI aims to cultivate a national culture of open science that maximizes the societal and economic impact of research and innovation, while reinforcing Qatar's position as a global hub for scientific excellence and collaboration.

1.2. Availability of the 2021 Recommendation in National Language(s)

The availability of the 2021 UNESCO Recommendation on Open Science in Arabic is a significant milestone for Qatar. Having this document in the national language, accessible through the provided link (https://unesdoc.unesco.org/ark:/48223/pf0000379949_ara), allows researchers, policymakers, educators, and the general public in Qatar to engage directly with its principles and guidelines, free from language barriers. This accessibility highlights UNESCO's commitment to inclusivity and the active participation of diverse stakeholders in the global open science movement. With the Recommendation available

in Arabic, Qatar can more effectively incorporate these principles into its national research strategies, educational curriculums, and innovation policies. This can lead to enhanced collaboration among researchers domestically and internationally, increased visibility and impact of Qatari research, and the promotion of a more transparent and equitable scientific ecosystem. Additionally, having the document in Arabic facilitates discussions and policy development at the national level, ensuring that the unique context and needs of Qatar's research and education landscape are taken into account during the implementation of open science practices. UNESCO's proactive step in providing the Recommendation in multiple languages, including Arabic, demonstrates a commitment to making the benefits of open science truly global and accessible to everyone.

1.3. National Awareness-Raising Activities on Open Science: Key Elements, Benefits, and Challenges (by End of 2025)

Qatar is actively engaged in fostering a national understanding of open science. This section outlines the key awareness-raising activities scheduled or already implemented by the close of 2025, that address the fundamental aspects, advantages, and potential challenges of open science, as detailed below:

National Scientific Research and Innovation Week (by MoEHE)

The event, held from September 29 to October 3, aimed to enhance students research, investigation and innovation skills, instill a spirit of cooperation and competition among them, achieve excellence in scientific research, highlight students research projects that address contemporary problems and provide them with the necessary support, provide an opportunity for distinguished projects to participate in regional and international competitions, and cooperate and coordinate with partners in state institutions concerned with scientific research¹.

Springer Nature and QNL OA awareness events

Springer Nature's agreement with Qatar National Library enables authors affiliated with any institute of higher education or non-profit research institute within Qatar to receive funding for book processing charges (BPC) when publishing their book OA with Springer Nature².

¹<https://www.edu.gov.qa/en/Events/Details/193800>

² <https://www.springernature.com/gp/open-science/oa-agreements/qatar>

QRDI Knowledge Transfer Training Under Mumaken Program

These Training sessions that helps to bridge the knowledge gap between research and industry while strengthening Qatar's position as a global leader in innovation³.

Collaborative and Continuous Awareness-Raising Activities by the QRDI Council

In collaboration with national stakeholders, the QRDI Council has already undertaken and continues to plan a series of awareness-raising activities focused on open science and its key elements. For instance, and in recent years, QRDI has embedded the principles of open science into its strategic frameworks, funding policies, and national dialogue. In 2022, the Council revised its funding policies for scientific research and talent tracks, explicitly incorporating open science principles such as transparency, open access to research outputs, and the ethical sharing of data. These policies are publicly available in both English and Arabic, reflecting the Council's commitment to accessibility and stakeholder engagement.

To raise awareness and foster a shared understanding of open science—including its benefits (e.g., increased research visibility, accelerated innovation) and challenges (e.g., data privacy, infrastructure needs)—QRDI Council has:

- Organized national consultations and policy workshops with ministries, universities, research institutions, and private sector partners to discuss pathways for realization on QRDI2030 and NDS3-2030.
- Established an online platform that not only houses national policies and programs but also provides a repository of Qatar-based research publications, and a catalogue of available research infrastructure, promoting resource sharing and research collaboration.
- Contributed to policy dialogues and national strategy development that align with UNESCO's open science recommendations and support inclusive knowledge ecosystems.

Looking ahead, by the end of 2025, the QRDI Council plans to expand these efforts through:

- Targeted awareness campaigns on open science practices across various sectors.
- Capacity-building workshops and training sessions, through Mumakin program, for researchers and policy practitioners on data sharing, open-access publishing, and responsible research.
- Strategic partnerships with international and regional organizations to promote harmonization with global open science standards.

³ <https://qrdi.org.qa/en-US/News-Details/Id/170>

1.4. Incorporating Open Science Values and Principles in Publicly Funded Research: National actions (by end of 2025)

Demonstrating a commitment to open science, Qatar is actively working to incorporate open science values and principles into publicly funded research endeavors. The following details the national actions being implemented:

Qatar University encourages research collaboration, transparency and knowledge dissemination. Open access publishing is available and encouraged through the colleges' scholarly journals⁴.

Qatar National Library is committed to free, open, unrestricted access to their collections and services⁵. Their values are: Openness and Transparency, Education and Lifelong Learning, Open Access, Equity, Diversity, Inclusion, Accessibility, Respect and Fairness, Growth and Innovation, Stewardship, and Intellectual Freedom.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1. National Policy, Strategy, or Plan of Action on science, technology and innovation (STI)

This section addresses the existence of a national policy, strategy, or plan of action for science, technology, and innovation (STI) in Qatar. Understanding the overarching framework for STI is crucial for contextualizing national efforts in areas like open science. The primary guiding framework in this regard is the Qatar National Vision 2030, which serves as the comprehensive national strategy directing the country's approach to science, technology, and innovation (STI).

The initial development of the Qatar National Vision 2030 was led by the General Secretariat for Development Planning (GSDP). It was indeed tasked with formulating the QNV 2030 framework.

⁴ <https://www.qu.edu.qa/en-us/qupress/Pages/journals.aspx>

⁵ <https://www.qnl.qa/en/about/vision-and-mission>

According to the National Planning Council⁶, the Qatar National Vision 2030 was formulated in 2008 to set the foundation for sustainable economic, social, culture and environmental development in Qatar. The development process of the National Development Strategy was launched in early 2010. The development process was carried out by various task teams representing all segments and sectors of Qatari society, under the supervision of 6 Executive Groups. 14 Sectors Strategies were completed in October 2010. The National Development Strategy is officially launched in March 2011 accompanied by extensive public communication efforts. Implementation of the National Development Strategy started after its official launch.

The vision focuses on four key pillars which are:

1. **Human Development:** This pillar focuses on building a strong human capital base to enable them to build a prosperous society.
2. **Social Development:** This pillar aims to create a secure society founded on strong moral values and social cohesion and promoting education, healthcare, and community well-being with the support of Science, Technology, and Innovation.
3. **Economic Development:** This pillar seeks to achieve sustainable economic growth, and it focuses on developing a knowledge-based economy.
4. **Environmental Development:** It highlights the significance of environmental sustainability and the protection of natural resources for future generations.

Though not explicitly using "open science", Qatar National Vision 2030 embodies its essential principles—open access, infrastructure, engagement, and dialogue—by emphasizing innovation, research, and international collaboration.

- **Open Access to Scientific Knowledge:** Qatar National Vision 2030 emphasizes education and a knowledge-based economy, which supports open access to scientific knowledge. Qatar's substantial investments in R&D infrastructure demonstrate a commitment to generating and sharing research.

⁶ <https://www.npc.qa/en/planning/Pages/nds.aspx>

- **Open Science Infrastructure:** It facilitates the growth of technological infrastructure, research capabilities, and partnerships with global research networks, which are interpreted as supporting the development of open science infrastructure, such as research repositories and data sharing platforms, even though this is not explicitly stated.
- **Open Engagement of Societal Actors:** The Vision 2030 emphasizes societal development and stakeholder involvement, fostering open engagement in science. Its focus on international collaboration further highlights this principle by integrating with the global scientific community.
- **Open Dialogue with Other Knowledge Systems:** The vision fosters international collaboration and knowledge exchange while preserving national heritage. It aligns with open science principles by encouraging cross-border knowledge sharing and dialogue among diverse knowledge systems. Furthermore, the QNV 2030's promotion of tolerance and understanding among different cultures and perspectives can be interpreted as supporting open dialogue with various knowledge systems, including traditional and indigenous knowledge.

While the Qatar National Vision 2030 provides the overarching framework for national development, including STI, the Qatar Research, Development, and Innovation (QRDI) Strategy offers a more detailed roadmap for achieving these ambitions. It acts as the primary mechanism through which Qatar's efforts in science, technology, and innovation are coordinated and advanced.

The Qatar Research, Development, and Innovation (QRDI) 2030 Strategy was adopted in 2020 and is spearheaded by the QRDI Council.

Key areas of the Strategy:

- National RDI Priority Areas (e.g., energy, environment, health, digital technologies and Society)
- Development and attraction of research and innovation talent
- Strengthening of innovation and commercialization pipelines
- Governance and coordination of the national RDI ecosystem
- Investment in research infrastructure and enabling regulations

Inclusion of Open Science Elements

The QRDI 2030 Strategy demonstrates a strong commitment to fostering an open, collaborative, and inclusive RDI environment. Several core elements of open science are explicitly supported in the strategy:

- **Open Access to Scientific Knowledge:** The strategy promotes the open dissemination of publicly funded research. It encourages researchers to publish in open-access journals and supports national platforms for research visibility.
- **Open Science Infrastructure:** Significant emphasis is placed on establishing digital research repositories, shared databases, and collaborative platforms that allow researchers, institutions, and the public to access and contribute to scientific knowledge.
- **Open Engagement of Societal Actors:** The strategy includes mechanisms to engage citizens and communities in innovation challenges, public science dialogues, and awareness initiatives. Societal participation is viewed as essential to driving demand-led research and fostering inclusive innovation.
- **Open Dialogue with Other Knowledge Systems:** Qatar's RDI ecosystem is positioned to promote international research collaboration and intercultural knowledge exchange. The QRDI 2030 Strategy encourages partnerships that bridge academic research, traditional knowledge, and global best practices. To support this vision, QRDI has established a number of concrete initiatives that foster global collaboration:
 - **International Joint Funding Programs:** QRDI has launched specific funding programs in partnership with international funding agencies. To date, collaborations have included countries such as Turkey and Spain, among others, enabling joint calls and co-funded research projects that address shared challenges.
 - **Global Access to Funding Opportunities:** Several of QRDI's research funding programs are open to international scholars and institutions, allowing researchers from outside Qatar to collaborate with local entities and contribute to national research priorities.

Strategic Partnerships in Research and Innovation: QRDI has signed formal research and innovation partnerships with a growing number of countries, including Singapore, Sweden, and the United Kingdom. These agreements are designed to facilitate knowledge exchange, joint initiatives, and long-term cooperation in areas of mutual interest.

- **National RDI Communities:** To advance science and nurture an open, knowledge-sharing society, QRDI cultivates Research, Development, and Innovation (RDI) communities aligned with national priority areas. These communities not only drive progress within their domains but also promote public engagement by openly sharing insights, discoveries, and innovations.

2.2. Existence of National Policies Aligning with the 2021 Recommendation that address Open Science and its elements

Qatar has several policies, strategies, and initiatives that address key elements of open science, aligning with the 2021 UNESCO Recommendation. The principles and key elements of open science, as outlined in the 2021 UNESCO Recommendation, are increasingly being addressed and integrated within the nation's broader research and innovation ecosystem. This integration is primarily driven by strategic initiatives and policies championed by key national entities, aiming to foster a more transparent, collaborative, and accessible research environment. The following points will delve into how these existing frameworks and ongoing efforts align with the core tenets of the UNESCO Recommendation.

SUB-POINT 16.a: There aren't detailed guidelines on how to ensure reciprocity and prevent inequitable extraction in the context of open science. However, the principles underlying 16(a) can be reflected in broader Qatari policies and practices. For example, Qatar's research funding bodies, like the QNRF, emphasize ethical research conduct. This includes respecting intellectual property rights and ensuring fair and equitable access to research benefits. These ethical considerations help prevent the unfair extraction of data and knowledge. Furthermore, there are laws and regulations concerning data privacy and security in Qatar. These regulations contribute to a framework where data sharing must be done responsibly and ethically. Finally, the Focus on National Development implies that any open science initiatives would need to serve Qatar's interests and wouldn't involve the unfair or inequitable extraction of its resources or knowledge.

SUB-POINT 16.b: Although Qatar does not have a specific policy mandating that all publicly funded research outputs (publications, data, software, etc.) must follow open licensing or be dedicated to the public domain, there are several policies and practices that suggest a tendency toward these principles, albeit with some nuances. Qatar promotes open access to publicly funded research, notably through QNRF and QNL initiatives as follows:

- The Qatar National Research Fund (QNRF) encourages open access to research outputs from funded projects. While it is not always a mandatory requirement, it represents a significant initiative to make publicly funded research openly available. Its policies address intellectual property rights, often including provisions that allow researchers to retain certain rights while also facilitating public access to research findings. This indicates a trend towards open licensing, although it may not always be the default option.
- The Qatar National Library open access program⁷ offers researchers several ways to publish their work openly. This includes funding for article processing charges and a repository for archiving research. By providing these resources, the program facilitates open access to publications resulting from research, which may include projects that are publicly funded.

SUB-POINT 16.c: Qatar's policies don't directly address "bibliodiversity" in the way described in UNESCO recommendation. There isn't a national policy specifically promoting diverse formats, community-driven publishing models, or prioritizing not-for-profit publishing. However, some elements contribute indirectly. The QNRF, along with universities such as Qatar University and Qatar Foundation, funds research in various disciplines. Although this funding does not specifically target publishing models, it supports the development of diverse research outputs. Furthermore, Qatar University Library (QUL) enhances access to a variety of publications and supports academic publishing by offering platforms like an institutional repository and an archive for open-access resources. Although it does not explicitly focus on bibliodiversity, QUL's services contribute to increasing access to diverse formats and types of knowledge.

SUB-POINT 16.d: There is no specific national policy that highlights the importance of multilingualism in scientific publications or academic communications. However, multilingualism is present in Qatar's academic and research environment due to the use of Arabic and English as follows: Arabic is the official language of Qatar and is promoted in education, government, and media. This emphasis extends to academic and scientific communication within the country. Qatar places a strong focus on education, including

⁷ <https://www.qnl.qa/en/open-access-at-qnl/about>

higher education and research, which fosters multilingualism. As a result, students and researchers are often exposed to English and other languages in their studies and professional activities. Consequently, many scientific publications from Qatari institutions are published in English. Additionally, Qatar's active participation in international research collaborations necessitates the use of multiple languages, particularly English, for scientific communication and publications.

SUB-POINT 16.e: Qatar is a member of the United Nations and has likely endorsed the UN Declaration on the Rights of Indigenous Peoples. However, this general endorsement may does not result in specific national legislation or policies that directly address the intersection of indigenous rights and open science practices. Nevertheless, future global collaborations may tackle these issues.

SUB-POINT 16.h: Qatar is actively developing a research and innovation ecosystem that brings together various stakeholders, including researchers, academics, and industry representatives. While addressing open science issues may not be their primary focus, this collaborative environment has the potential to contribute to those discussions. Institutions like Qatar University (QU) host conferences and workshops on research, innovation, and open access, providing opportunities for multi-stakeholder dialogues on open science, even if they do not specifically address the challenges outlined in section 16(h). Additionally, the Qatari government sometimes conducts public consultations on policy matters, creating a platform for gathering diverse perspectives that could also be utilized for discussions on open science.

2.3. Dedicated National Policy Instruments for the Promotion of Open Science

Qatar is making efforts to promote open science, but it is still in the early stages of establishing formal policy instruments for this purpose. However, Qatar has policy instruments within its broader science, technology, and innovation (STI) framework that promote open science by enabling open access to research, fostering collaboration, and supporting transparency. Some of the key policies and initiatives that are contributing to the advancement of open science in Qatar include:

1. Qatar National Research Strategy (QNRS)

The main objective of Qatar National Research Strategy (QNRS) is to guide and prioritize research efforts to support Qatar's national development goals and its transition to a knowledge-based economy.⁸

Elements of open science included:

The Qatar National Research Strategy (QNRS) is essential to Qatar's research and innovation framework, promoting open science practices in accordance with UNESCO's recommendations. The components of open science within QNRS include:

- **Open Access to Scientific Knowledge:** QNRS promotes open access publishing, ensuring publicly funded research is available to everyone, in line with UNESCO's objectives for open access to scientific knowledge.
- **Open Science Infrastructure:** The strategy promotes the development of infrastructure, including digital repositories and data-sharing platforms, to facilitate open access to research outputs.
- **Open Engagement of Societal Actors:** The QNRS highlights the importance of collaboration among universities, industries, and government agencies, fostering transparency and allowing diverse societal participants to engage in research.
- **Open Dialogue with Other Knowledge Systems:** it enhances international collaboration through research networks, facilitating the exchange of scientific knowledge and fostering an inclusive, diverse research ecosystem.

2. Qatar National Research Fund (QNRF)

The main objective of Qatar National Research Fund (QNRF) is fostering original, competitively selected research in engineering and technology, physical and life sciences, medicine, humanities, social sciences, and the arts.⁹

⁸ <https://www.mozabintnasser.qa/en/news/qatar-national-research-strategy-heralds-new-era-innovation#:~:text=The%20QNRS%20was%20developed%20to,to%20align%20objectives%20with%20national>

⁹ <https://www.qf.org.qa/research/qatar-national-research-fund>

Elements of open science included:

The Qatar National Research Fund (QNRF), under the Qatar Foundation, plays a crucial role in promoting open science in Qatar through its research funding programs. These programs encourage projects that adhere to open science principles, particularly through the National Priorities Research Program (NPRP), which urges researchers to make their findings publicly accessible. The NPRP prioritizes research that has significant societal impact and aligns with national goals. Additionally, QNRF's support for open data initiatives allows researchers to share their data and results widely, promoting transparency and collaboration both within Qatar and internationally. QNRF's funding models also support open science policies by encouraging open access publishing and data sharing whenever possible.

Key elements of open science in QNRF's programs include:

- **Open Access to Scientific Knowledge:** Projects funded by QNRF are encouraged to adopt open access policies, ensuring the free dissemination of research outcomes such as publications and data to the global community.
- **Open Science Infrastructure:** It facilitates the creation of platforms for storing, sharing, and accessing data and research findings, supporting the development of open science infrastructure.
- **Open Engagement of Societal Actors:** It promotes collaborations among researchers, policymakers, and industry representatives to ensure that research effectively addresses societal challenges and engages the public.
- **Open Dialogue with Other Knowledge Systems:** it fosters global collaborations and partnerships that facilitate cross-border knowledge exchange and sharing, thus contributing to international open science initiatives.

3. Qatar University Initiatives and policies

As a leading academic institution, Qatar University plays a vital role in fostering open science. This is evident in its initiatives, such as the Qatar University Institutional Repository (QSpace), which provides a platform for open access to research.

Furthermore, the university actively promotes open science by hosting conferences and workshops that address open access, open data, and other related topics. These activities create valuable opportunities for collaboration and dialogue, contributing significantly to the advancement of open science in Qatar.

The main objective of Qatar University institutional Repository (QSpace) is to collect materials originating from faculty, students, and researchers of Qatar University, provide necessary metadata to fulfill a complete bibliographic record, provide open access full-text content to everyone, and preserve content in a secure format for long-term continued access.¹⁰

Elements of open science included:

QSpace, the institutional repository of Qatar University is fundamental component of Qatar's open science infrastructure. It facilitates open science practices in the following ways:

- **Open Access to Scientific Knowledge:** QSpace provides open access to academic research, theses, dissertations, and publications produced by Qatar University's faculty and students, ensuring that scholarly works are visible to a wider audience.
- **Open Science Infrastructure:** The repository serves as a centralized digital platform for storing, preserving, and sharing research data and scholarly publications, contributing to Qatar's open science infrastructure.
- **Open Engagement of Societal Actors:** By making research publicly accessible, QSpace fosters engagement with diverse societal stakeholders. This allows the general public, industry professionals, and academic researchers to both benefit from and contribute to Qatar's research efforts.
- **Open Dialogue with Other Knowledge Systems:** By providing open access to research at Qatar University, this initiative fosters global visibility and collaboration. It encourages interdisciplinary exchanges, promotes regional knowledge sharing,

¹⁰ <https://qspace.qu.edu.qa/handle/10576/20491>

and promotes open dialogues with other knowledge systems around the world, aligning with the goal of inclusive global scientific collaboration.

4. Qatar Digital Library (QDL)

Qatar Digital Library is developed as part of a longstanding partnership between Qatar Foundation, Qatar National Library and the British Library.¹¹

The main objective is to provide free online access to a wealth of historical source materials about the Middle East, with a particular focus on the Gulf region, thereby promoting knowledge and understanding of the region's history, culture, and heritage.¹²

Elements of open science included:

The Qatar Digital Library is a crucial component of Qatar's open science strategy. These initiatives align with the UNESCO Recommendation on Open Science in several ways:

- **Open Access to Scientific Knowledge:** It offers free and open access to a vast collection of historical, cultural, and scientific resources, including manuscripts, maps, and archival materials, promoting global access to knowledge.
- **Open Science Infrastructure:** It is a key component of Qatar's open science infrastructure, facilitating the storage, sharing, and dissemination of knowledge through digital platforms.
- **Open Engagement of Societal Actors:** It engages a diverse range of societal actors, including researchers, students, and the general public, by providing resources that enhance education and scholarly activities.
- **Open Dialogue with Other Knowledge Systems:** QDL's international collaboration and accessible resources promote dialogue among various cultural and academic knowledge systems, enabling cross-border and interdisciplinary exchanges.

¹¹ <https://www.qdl.qa/en/about>

¹² <https://www.qnl.qa/en/about/news/qatar-national-library-and-british-library-celebrate-qatar-digital-librarys-10th#:~:text=Launched%20by%20Qatar%20National%20Library,historical%20and%20cultural%20heritage%20through>

5. QRDI 2030 Strategy and the QRDI Data Management Policy

The objective of the QRDI 2030 Strategy is to drive research, development, and innovation efforts in Qatar in alignment with the Qatar National Vision 2030. While, the objective of the QRDI Data Management Policy is to establish guidelines and principles for the management of research data generated by projects funded by the QRDI Council.

Elements of open science included:

Qatar has specific policy instruments that aim to promote open science, primarily through the QRDI 2030 Strategy and the QRDI Data Management Policy. While there is no standalone national open science policy yet, these instruments collectively embed open science principles into the country's research and innovation ecosystem. The QRDI 2030 Strategy emphasizes open access to publicly funded research, international collaboration, and inclusive knowledge exchange. Complementing this, the QRDI Data Management Policy requires researchers funded by QRDI to adopt responsible data practices, including the development of data management plans and adherence to the FAIR principles—ensuring data is Findable, Accessible, Interoperable, and Reusable. It also promotes the sharing of research data and outputs, while safeguarding ethical, legal, and privacy considerations. Together, these instruments advance transparency, accessibility, and global engagement in scientific research, reflecting Qatar's growing commitment to open science.

6. Innolight

The main objective of Innolight is to be a comprehensive national online platform that connects key stakeholders in Qatar's Research, Development, and Innovation (RDI) ecosystem¹³.

¹³ <https://innolight.qrdi.org.qa/communities>

Elements of open science included:

Innolight, Qatar's national platform for RDI knowledge exchange and collaboration, can make significant contributions to key aspects of open science as recommended by the 2021 UNESCO Recommendation:

- **Open Access to Scientific Knowledge:** It strives to make insights, experts, projects, facilities, and resources readily accessible¹⁴. By centralizing information about research outputs, expertise, and infrastructure, it facilitates the discovery and potential reuse of scientific knowledge. This approach aligns with the principles of open access to publications, data, and potentially even resources and software.
- **Open Science Infrastructure:** It functions as a digital open science infrastructure. It offers a platform for sharing information, connecting stakeholders, and potentially hosting or linking to open access repositories, databases, and collaborative tools¹⁵. This aligns with the Recommendation's emphasis on investing in and developing open science infrastructures.
- **Open Engagement of Societal Actors:** It promotes greater participation in the scientific process by connecting innovators, researchers, industries, and institutions¹⁶. It encourages dialogue and collaboration among various stakeholder groups, including those outside the traditional scientific community, which is a crucial aspect of open engagement.
- **Open Dialogue with Other Knowledge Systems:** Innolight's role in connecting various stakeholders could help foster dialogue and collaboration between holders of conventional scientific knowledge and those with indigenous or traditional knowledge.

2.4. National Roadmap, policy or plan for Implementing the 2021 Recommendation on Open Science

Qatar has been developing a comprehensive, national-level roadmap or policy aimed specifically at implementing open science, as recommended by the 2021 UNESCO

¹⁴ <https://innolight.qrdi.org.qa/insights>

¹⁵ <https://innolight.qrdi.org.qa/>

¹⁶ <https://thepeninsulaqatar.com/article/25/02/2025/qrdi-council-launches-online-platform-innolight>

Recommendation on Open Science. Discussions and planning for a national open science strategy may be underway in Qatar as awareness of the benefits of open science continues to grow. It is likely that Qatar will further develop its approach to implementing the UNESCO recommendations. In the meantime, existing initiatives provide a solid foundation for advancing open science in Qatar. By continuing to support these efforts and encouraging dialogue among stakeholders, Qatar can make significant progress toward creating a more open and accessible research environment.

The significant efforts, several initiatives and frameworks in Qatar support the principles of open science include:

- **Qatar Open Data Portal:**

This open portal is a national data platform that indexes, tabulates, disseminates, analyzes and visualizes official statistics to facilitate their use. It should be noted that the data available on this platform are disseminated in full cooperation with partners' sources and can be circulated and used by all users of different interests, to be fully utilized.¹⁷

Qatar's Open Data Portal is a crucial component of Qatar's open science strategy. These initiatives align with the UNESCO Recommendation on Open Science in several important ways:

- Encourages the sharing of publicly available data, which promotes open science practices by ensuring transparency and the sharing of knowledge.
- Contributes to the creation of infrastructure for open science, providing the public with access to a range of governmental data.
- Makes information accessible to the public, fostering greater societal engagement with data, and encourages innovation in research, development, and policy-making to help address societal challenges, as recommended by UNESCO.

¹⁷ <https://www.data.gov.qa/pages/default/>

- Ensures data is accessible to various stakeholders, such as researchers, policymakers, and the general public. This promotes inclusivity and democratizes access to information, which is a fundamental aspect of open science.
- Facilitates global collaboration and knowledge exchange, helping integrate global knowledge systems into local research efforts.

- **The Third National Development Strategy**

The Third National Development Strategy (NDS-3) aims to foster a knowledge-based economy, develop human capital, promote digital transformation, encourage international collaboration, and drive innovation in key sectors. These overarching goals create a robust environment for implementing open science principles at the national level, aligning closely with the objectives of the 2021 UNESCO Recommendation. The focus on research and development, technology adoption, and creating a collaborative ecosystem within NDS-3 establishes a strategic framework in which open science initiatives can flourish and contribute to Qatar's national development goals¹⁸.

Additionally, libraries are essential in advancing national strategies for open science in accordance with the 2021 UNESCO Recommendation on Open Science.

- **Qatar University Library (QUL):** QUL is a strong supporter of open science and open access, offering resources that align with UNESCO recommendations. Its commitment is evident through various services and initiatives that contribute to the broader open science landscape. These include:
 - QSpace, the institutional repository that aligns with UNESCO's recommendations for open access to scholarly works at Qatar University. This platform allows faculty and researchers to deposit and share their research outputs easily.

¹⁸<https://www.npc.qa/en/planning/nds3/Pages/default.aspx>

- Additionally, QUL actively encourages open access publishing by providing guidance to researchers about available options.
- The library also collaborates with other institutions to advance open science, offering resources to help researchers in research data management, which is a vital aspect of open science.

QUL plays a key role in supporting open science at Qatar University by providing infrastructure and resources that foster a culture of open science among faculty and researchers. Its active support for open access and research data management demonstrates a strong commitment to the principles outlined in the 2021 UNESCO recommendations.

- **Qatar National Library (QNL):** QNL's actions and initiatives demonstrate a strong commitment to open science principles through several key initiatives.
 - Its Open Access Program provides support for researchers to make their publications freely available. This includes establishing open access agreements with publishers, implementing an article processing charge (APC) fund, promoting alternative publishing models, and participating in international initiatives.
 - The Manara digital repository archives and offers access to research conducted in Qatar, aligning with UNESCO's recommendations for open access to scholarly information by enabling researchers to deposit and share their outputs.

2.5. Institutional Policies and Strategies Promoting Open Science

1. Qatar University Institutional Repository Policy

Qatar University has made a notable advancement in promoting open science principles by developing and implementing its Institutional Repository Policy, in line with the 2021 UNESCO Recommendation. This policy provides a framework for the systematic

collection, preservation, and open dissemination of the university's diverse scholarly outputs. As a result, it enhances the visibility, accessibility, and impact of its research and academic work both within Qatar and globally.

Qatar University has demonstrated its commitment to open science through its Institutional Repository Policy. It is now essential to explore how this policy aligns with the key elements of open science outlined in the 2021 UNESCO Recommendation. This discussion will examine how the QSpace policy facilitates open access, supports data sharing, encourages collaboration, and promotes other important aspects of open science within the university's research and academic community.

Addressing Elements of Open Science:

The Qatar University Institutional Repository (QSpace) Policy serves as an example of a policy that promotes open science at the institutional level and aligns with various elements of the 2021 UNESCO Recommendation on Open Science. Although this policy focuses specifically on the university, its alignment with key aspects of open science indicates that Qatar University is actively contributing to national open science practices.

QSpace Policy is aligned with the specific recommendations in section 17 (ii) of the UNESCO 2021 Recommendation on Open Science as follows:

- Promoting Open Science at the Institutional Level (17.d)

The Qatar University Institutional Repository Policy is essential for promoting open science practices at Qatar University. It ensures that research outputs created by faculty, researchers, and staff are readily accessible to the public. This aligns with sub-point 17.d, which highlights the responsibility of research-performing institutions and universities to support open science principles. The policy mandates that various research outputs—including faculty contributions, student theses, and research publications—are made available in digital format. This contributes to open access and facilitates the broader dissemination of knowledge.

By providing a way for users to request restricted documents, this feature helps overcome paywalls and copyright barriers, ensuring that valuable research remains accessible to those with a legitimate need, thus promoting a more inclusive open-access model.

- Support for Open Science in Universities and Research Performing Institutions (17.e)

The QSpace policy promotes the idea that universities and research-performing institutions should take an active role in fostering open science. It aligns with sub-point 17.e, which advocates for the establishment of policies, strategies, and infrastructures that create conditions conducive to open science. Qatar University's initiatives to develop and maintain an institutional repository significantly contribute to building a supportive infrastructure for open science practices. The policy offers a structured framework for the collection, storage, and dissemination of research outputs, all of which support open science principles. It also enables self-depositing, allowing faculties and researchers to upload their work to QSpace. This ensures that the dissemination of knowledge is decentralized and accessible to the broader academic community.

- Encouraging Collaborative Open Science Practices (17.g)

This policy encourages interdisciplinary collaboration by making research accessible to other researchers, not only within Qatar but also across global academic communities. As a result, it fosters collaborative partnerships among researchers and encourages participation in global open science initiatives. QSpace serves as a platform that facilitates collaboration across various institutions by providing easy access to publicly available research, thereby contributing to open science discussions both within and beyond Qatar's academic community. Furthermore, since QSpace encourages researchers to self-deposit their work, it enables them to freely share their research. This approach simplifies engagement for academic unions and societies, allowing them to engage with and benefit from this open repository of knowledge.

To effectively address all elements of open science, including citizen and participatory science and the co-production of knowledge among multiple actors, we must expand our focus beyond institutional repositories like QSpace. A more comprehensive framework is needed to integrate open science practices across various levels of society and research. This involves developing policies and strategies that engage academic institutions, citizens, public organizations, and private sector stakeholders. Such integration could be achieved through national-level frameworks that promote open access, citizen science, public engagement, co-production of knowledge, and multi-actor collaborations in a holistic way.

2. Policies and institutional strategies Guided by the QRDI 2030 Strategy

In Qatar, there are emerging policies and institutional strategies that promote open science in alignment with the 2021 UNESCO Recommendation, particularly within research-performing institutions and universities. Guided by the QRDI 2030 Strategy and

the national push for responsible research practices, institutions are increasingly embedding open science principles into their research governance frameworks. For example, universities and research centers receiving QRDI funding are required to submit data management plans that adhere to open access and FAIR data principles. The QRDI Council's Data Management Policy serves as a key reference for institutional compliance, encouraging the sharing of research outputs while protecting data privacy and ethical standards. Additionally, several universities have adopted or are in the process of developing open access policies, establishing institutional repositories, and fostering collaborations that support open and inclusive research. These efforts reflect a growing institutional alignment with global standards for open science in Qatar's academic and research ecosystem.

2.6. National Funding Mechanisms for Open Science

Qatar has various funding mechanisms that support open science, although open science is often integrated into broader research funding initiatives rather than being a standalone focus. Below are some of the key mechanisms:

Qatar Research, development and Innovation (QRDI): it has established a robust suite of national-level funding mechanisms that promote open science, administered through the Council and accessible via the Innolight platform. These mechanisms are structured under three main tracks: Scientific Research, Innovation, and Talent and Capability Building. Under the Scientific Research track, programs such as the Academic Research Grant (ARG), Academia-Industry Cooperation Call (AICC), Climate Change and Environment Call (CCEC), and Path Towards Precision Medicine (PPM) support high-impact, open, and collaborative research aligned with national priorities. The Innovation track features programs like Qatar Open Innovation (QOI), Technology Development Grant (TDG), Qatar-Spain Innovation Program (QASIP), and the Knowledge Transfer Partnership Qatar (KTP-Q), all of which emphasize industry-academia collaboration and the open exchange of solutions. Meanwhile, the Talent and Capability Building track includes initiatives like the Graduate Student Research Award (GSRA), Undergraduate Research Experience Program (UREP), Displaced Arab Researcher (DAR), and the Technology Innovation Training Program, which foster a new generation of researchers committed to openness and knowledge sharing. These funding programs frequently incorporate open science principles—such as open access, international collaboration, and data transparency—reflecting Qatar's growing commitment to embedding open science across its research and innovation landscape.

Qatar National Research Fund (QNRF): It is the primary national funding agency for research in Qatar. It provides significant research funding across various fields. While its primary focus is on advancing scientific research and it doesn't have a separate open science funding track, its policies and priorities encourage practices such as open access and research data sharing, which align with open science principles.

- It requires researchers to include data management plans in their grant proposals, which should outline provisions for data sharing and open access.
- It encourages researchers to publish their findings in open access journals or repositories.
- It emphasizes on research areas that benefit society and promote open science by funding projects involving community engagement.

Qatar University funding: Qatar University offers internal funding mechanisms to support open science activities by providing funds for researchers who publish in open access journals and by investing in its institutional repository, QSpace, which provides infrastructure for open access to research outputs.

Collaboration with International Funding Bodies: Qatar is engaged in funding partnerships with the European Union and various research networks. These collaborations often provide funding opportunities that promote open science practices.

Qatar's funding mechanisms for open science are developing, with increased targeted opportunities anticipated in the future. Researchers in Qatar can explore funding options from organizations such as the Qatar National Research Fund (QNRF), the Qatar Research, Development and Innovation (QRDI), and their respective universities to support projects that incorporate open science practices. By emphasizing the open science components of their research in grant proposals, researchers can enhance their chances of securing funding.

2.7. Integration of Open Science Practices in Existing Research Funding

Open science practices have been integrated into existing research funding mechanisms, particularly through the major funding body supporting scientific research in the country, For example QNRF. These practices are being integrated in several key ways:

- **Encouraging Open Access:** Researchers who receive QNRF funding are often required that their research findings publicly available through open-access platforms.

- **Data Sharing:** Encourages sharing research data openly and ensures that research data is accessible to other researchers to promote transparency and reproducibility which are the core aspects of open science.
- **Support for Collaborative Research:** Encourages multi-disciplinary, cross-institutional, and international research collaborations, involving both public and private sector stakeholders, to create an open research environment.

In summary, significant progress has been made in promoting open access, data sharing, and collaborative research within Qatar's research funding landscape, indicating a growing integration of open science principles. Development of dedicated open science funding would incentivize and support researchers in adopting open science practices. Funding agencies could further incentivize open science by integrating it into grant evaluations and rewarding researchers for their active engagement in open science. Funding should be accompanied by training and resources to provide researchers with the skills and knowledge necessary to effectively implement open science practices.

2.8. Efforts towards fostering Equitable Public-Private Partnerships on Open Science

Qatar is actively fostering efforts towards the equitable public-private partnerships (PPPs) in open science, with potential directions and areas of focus emerging from current trends and the existing research and innovation landscape. The points below highlight these potential directions and areas of focus:

- QRDI, a key player in Qatar's research and innovation, will likely be involved in promoting public-private partnerships on open science through funding, policy, or facilitating public-private connections.
- Public-private partnerships on open science will likely align with Qatar's national priorities (e.g., Qatar National Vision 2030), focusing on impactful sectors like health, environment, and technology.
- Existing Qatari open access and data management initiatives provide a foundation for developing open science PPPs, potentially through expanded private sector participation.
- Qatar's strong focus on academia-industry-government collaboration, coupled with its existing international research collaborations, suggests a continued prioritization and potential modeling for national public-private partnerships in open science.

- The Qatar Foundation (QF) Innovation Program, primarily through Qatar Science & Technology Park (QSTP)¹⁹, fosters collaboration among academia, government, and industry. By offering essential infrastructure, funding opportunities, and platforms for engagement, it establishes a foundation for equitable public-private partnerships (PPPs) in open science. This approach enables the sharing of knowledge and resources to drive innovation and benefit society²⁰, in line with the principles of the 2021 UNESCO Recommendation.
- The operational model of the Qatar Biobank²¹, along with its collaborative nature²² and alignment with national strategies, highlights its significant role in promoting partnerships that advance medical research²³ and innovation in Qatar. By focusing on data sharing within ethical guidelines and facilitating collaborative research, the Qatar Biobank serves as a key enabler for open science initiatives that engage both public and private stakeholders.
- Sidra Medicine, Sidra Medicine collaborates with the Qatar Research, Development, and Innovation (QRDI) Council on initiatives such as the Qatar Open Innovation (QOI) program²⁴. These programs encourage startups, small and medium-sized enterprises (SMEs), and corporations to propose innovative solutions to specific challenges within the healthcare sector. This collaboration creates direct opportunities for public-private partnerships (PPPs) aimed at developing and implementing new technologies²⁵.

2.9. National Monitoring Framework or Indicators for Open Science

Qatar has demonstrated significant advancement in open science, with indications that a dedicated national monitoring framework, potentially including specific open science indicators, either exists at some level or is currently under development. This progress is reflected in the following ways:

¹⁹ <https://qstp.org.qa/>

²⁰ <https://www2.fundsforngos.org/donor-agencies/qatar-foundation-advancing-education-and-innovation-in-qatar/>

²¹ https://www.qphi.org.qa/sites/default/files/2024-05/General%20Leaflet_en.pdf

²² https://www.researchgate.net/figure/Qatar-Biobank-operational-plan-collaboration-with-multidisciplinary-stakeholders_fig1_337933667

²³ <https://www.zawya.com/en/business/healthcare/qatar-biobank-makes-significant-contribution-to-local-health-research-cp25ns7p>

²⁴ <https://qrqi.org.qa/en-US/News-Details/Id/151>

²⁵ <https://www.sidra.org/media/news/2025/sidra-medicine-and-rasmal-ventures-partner-to-drive-digital-health-innovation-in-qatar/>

- Government bodies or institutions like the QU, QF, QNRF, QRDI, or QNL is working on tracking relevant metrics internally to assess the impact of their open science-related initiatives.
- Open science indicators may be embedded within broader research and innovation monitoring, such as metrics for data sharing, open access publications, or collaboration, even if not explicitly labeled as "open science" indicators.

From the perspective of the Qatar Research, Development, and Innovation Council (QRDI), while there is currently no standalone national monitoring framework dedicated exclusively to open science, the principles and components of open science are increasingly being integrated into broader national STI (Science, Technology, and Innovation) policy frameworks.

QRDI's 2030 Strategy and the ongoing implementation of Qatar's Third National Development Strategy (NDS-3), the Innovation track- emphasize open access to research outputs, collaboration, knowledge sharing, and data-driven innovation.

Furthermore, QRDI²⁶ actively promotes open science through policy dialogue, stakeholder engagement, and strategic partnerships with national and international entities. These efforts aim to develop clearer national guidelines and indicators for open science as part of the broader RDI monitoring framework.

Going forward, the integration of open science indicators into the national STI M&E framework is a priority and aligns with Qatar's commitment to responsible research, transparency, and global collaboration.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1. National Gross Domestic Product (GDP) Dedicated to R&D Expenditure

According to the latest data, Qatar has been steadily increasing its investment in research and development (R&D). The most recent figures from the World Bank indicate that Qatar's R&D expenditure as a percentage of GDP was 0.68% in 2021. Please refer to The World Bank: <https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS>

²⁶ <https://qrdi.org.qa/en-US/Strategic-Focus>

3.2. Percentage of Population with Reliable Internet Access and Bandwidth

As of the most recent data available, Qatar boasts one of the highest rates of internet access and broadband penetration globally. According to DataReportal's Digital 2024 report, 99.0% of the population in Qatar has internet access. For more information, you can refer to the DataReportal link²⁷.

The high level of connectivity in Qatar is the result of significant infrastructure investments and the government's efforts to create a digitally connected society. The Qatar National Vision 2030 emphasizes the importance of technology and digital infrastructure, with the goal of promoting a knowledge-based economy. This includes expanding internet access and enhancing digital services.

3.3. Active National Research and Education Networks (NRENs)

Qatar has a National Research and Education Network (NREN) called the Qatar National Research and Education Network (QNREN)²⁸. This network is crucial in supporting research and educational institutions by providing high-speed, reliable, and secure connectivity for research, academic, and collaborative purposes²⁹. QNREN is managed by Qatar University and facilitates access to global research infrastructure.

QNREN plays a crucial role in promoting open science in Qatar due to its accessibility and alignment with international research and education networks. It supports a collaborative research environment both regionally and globally. By linking various stakeholders, QNREN enables collaboration, data sharing, and access to worldwide research resources. These fosters open science practices and strengthens the overall research ecosystem in Qatar.

3.4. National and Institutional Open Science Infrastructures

²⁷ <https://datareportal.com/reports/digital-2024-qatar#:~:text=There%20were%202.70%20million%20internet,penetration%20stood%20at%2099.0%20percent>

²⁸ <https://asren.net/blog/qatar>

²⁹ <https://www.qu.edu.qa/en-us/Offices/its/project-management-office/Pages/projects-initiatives.aspx>

Open science infrastructure typically refers to systems and tools that support the sharing, dissemination, and open access to research outputs, data, and scholarly content. In Qatar, key examples of developing national and institutional open science infrastructure include:

1. Qatar University Institutional Repository (QSpace)

QSpace, as an open access repository, provides a platform for the university's faculties and researchers to share their work, making it publicly available and accessible. This aligns with key open science principles, such as transparency, accessibility, and collaboration.

QSpace is a digital archive that supports the dissemination of freely accessible knowledge which is a significant part of open science. by hosting a variety of academic outputs like research papers, theses and dissertations, and other scholarly materials. It serves as a platform where researchers can share their findings with more people, which in turn fosters collaboration and proliferation of knowledge.

Ideally, it should also come with additional functionalities like data sharing, collaboration tools, and open peer review support to be fully integrated into open science practices. Nevertheless, QSpace, which is a primary source of research outputs, certainly serves an important role in the open science movement.

a. QSpace Repository: Infrastructure Types

Having identified QSpace repository as a key example of open science infrastructure in Qatar, it is pertinent to examine the specific types of infrastructure exemplified by Qatar University's QSpace repository in relation to the categories outlined by UNESCO recommendations. The following discussion will explore how QSpace mainly fits within the categories of community-managed infrastructures and platforms for exchanges and co-creation of knowledge, shedding light on its role in advancing open science within the national context:

- **Community-managed infrastructures, protocols, and standards:** QSpace follows established protocols and standards for repository management (e.g., metadata standards, etc.). Moreover, it is a model of a community-managed infrastructure that supports the university's publication and dissemination of scholarly works.
- **Platforms for exchanges and co-creation of knowledge between scientists and society:** Making research outputs public, QSpace utilizes this as a

communicational medium for scientists and society. This is particularly relevant in terms of research that has practical applications or societal relevance. Researchers can reach out to other audiences apart from academic ones, thus, contributing to the quest for the general good of society.

b. QSpace Compliance with 2021 UNESCO Recommendation Principles (FAIR & CARE)

QSpace aligns with the 2021 Recommendation on Open Science by promoting inclusivity through open access, reducing barriers for diverse researchers and ensuring diverse language support for both content and interface.

It ensures accessibility with freely available research to a wider audience while respecting intellectual property rights. It is supporting interoperability by adopting standards for integration with other platforms, enabling seamless data exchange and collaboration.

Community governance is prioritized, allowing researchers, librarians, faculties and other stakeholders to collaboratively shape the platform's development to ensure the platform evolves to meet the needs of the community and remains responsive and accountable.

In line with the FAIR principles, QSpace ensures research is findable likely assigns persistent identifiers (like Handles and DOIs) to research outputs, making them easily discoverable. It ensures research is accessible by offering open access to many research outputs, with clear licensing information. It is interoperable, and reusable through proper metadata, clear permissions, and system compatibility.

It also adheres to the CARE principles by fostering collective benefit through contribute to the advancement of knowledge, giving researchers authority over their work, encouraging ethical research practices by incorporate ethical considerations into its platform design and policies, and it ensure responsible data use.

c. Non-Commercial Nature of QSpace Infrastructure

QSpace is a non-commercial Platform maintained by Qatar University Library, which means its primary goal is to support the university's academic community and promote open access to research. It focuses on supporting the sharing and dissemination of

knowledge for the public good which aligns with the UNESCO recommendation that advocates for infrastructures that are not driven by commercial interests.

2. Hamad bin Khalifa University Research Institutes

Hamad Bin Khalifa University (HBKU) research institutes—Qatar Computing Research Institute (QCRI)³⁰, Qatar Energy and Environment Research Institute (QEERI)³¹, and Qatar Biomedical Research Institute (QBRI)³² are key components of the open science ecosystem within HBKU, supporting transparency, collaboration, and the free flow of knowledge in both local and global contexts.

3. Manara:

It launched in 2022 by Qatar National Library, Manara is an open-access repository powered by Figshare, providing a platform for sharing research outputs³³.

3.5. Funding of Federated Regional or International IT Infrastructure for Open Science

Qatar is actively involved in supporting initiatives that contribute to the broader global open science landscape. The country's investments in digital infrastructure, open access initiatives, and research funding suggest a commitment to the principles of open science, which could contribute to broader efforts in this area. It's important to mention the context of Qatar's focus on digital infrastructure and open access, as well as related initiatives:

- The Qatar Research, Development, and Innovation (QRDI) Council, actively supports and engages in international collaborations that enhance access to research funds. While Qatar does not currently host a regional or global federated IT infrastructure dedicated solely to open science, QRDI has established strategic partnerships with international institutions and governments—including those in Singapore, Sweden, the UK, and Spain—that involve shared research platforms, data access, and digital collaboration tools. These partnerships enable Qatari researchers to participate in global open science initiatives and benefit from

³⁰ <https://www.hbku.edu.qa/ar/qcri>

³¹ <https://www.hbku.edu.qa/en/qeeri>

³² <https://www.hbku.edu.qa/ar/qbri>

³³ <https://manara.qnl.qa/>

federated infrastructure for data sharing, joint research, and open access publishing. Through funding programs such as the Qatar Open Innovation (QOI) platform, the Qatar-Spain Innovation Program (QASIP), and other bilateral collaborations, QRDI promotes cross-border research engagement and access to advanced digital environments that align with the principles of open science.

- Qatar funding research projects that prioritize data sharing and openness through QNRF which its funding aligns with international open science efforts by supporting collaborations that facilitate the sharing of scientific knowledge across borders.
- Qatar participates in regional and global efforts, such as through its contributions to platforms like DSpace, which has been used to establish the first institutional open-access repository, QSpace. It is managed by Qatar University Library which facilitates open science practices within Qatar and beyond.

3.6. Qatar's Involvement in North-South, South-South Collaborations for Open Science Platforms

Qatar is dedicated to promoting multinational partnerships, funding research with international components, supporting open science platforms, and optimizing infrastructure through various international collaborations. This aligns with the UNESCO 2021 Recommendation on Open Science, as evidenced by its participation in global and regional initiatives promoting knowledge sharing, open access, and collaborative research, potentially including infrastructure sharing and joint strategies.

QRDI actively engaged in North-South, North-South-South, and South-South collaborations to strengthen the use of shared research infrastructure and foster joint strategies for open science platforms. The has established bilateral and multilateral partnerships with nations such as Spain, Turkey, Singapore, Sweden, and the United Kingdom, supporting joint funding programs, knowledge exchange, and collaborative use of research infrastructure. For example, programs like the Qatar-Spain Innovation Program (QASIP) and partnerships under Qatar Open Innovation (QOI) create shared platforms for research and innovation that transcend national boundaries.

These collaborations not only optimize the use of physical and digital infrastructure but also promote capacity building, especially with countries from the Global South.

Furthermore, Qatar supports international scholars and displaced researchers—particularly through the Displaced Arab Researcher (DAR) program—illustrating a commitment to inclusive, cross-regional engagement. These efforts align with the principles of the UNESCO Recommendation on Open Science, particularly in advancing equitable access to infrastructure, fostering co-creation of knowledge, and building resilient research systems through global solidarity.

Hosting Forum for Open Research in MENA (FORM)³⁴ significantly contributes to North-South and South-South collaborations for open science platforms.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1. Systematic Capacity Building on Open Science in Higher Education and Research Institutions

Qatar is actively involved in initiatives that contribute to building capacity in open access, data sharing, and related open science principles through workshops, funding mechanisms, open access publishing platforms, and participation in regional forums. It is reasonable to expect that these efforts will continue to evolve and potentially become more formalized in the future, aligning with the global trend towards open science. The following outlines specific institutions and their programs dedicated to fostering open science practices and competencies in the higher education and research sectors:

Qatar University (QU):

- National capacity-building program that enables young researchers to develop their research portfolio, boost research productivity and conduct research that is shared in the Open Access repository (Qatar University institutional Repository).
- Continuing to provide support and create awareness of OER through a well-established office “Office for Digital Learning and Online Education”³⁵ at QU.

³⁴ <https://forumforopenresearch.com/about-form/>

³⁵ <https://www.qu.edu.qa/en-us/offices/dloe/about/>

- Al-Bairaq program is organized by the Qatar University Young Scientists Centre (QUYSC) and developed with UNESCO and the Qatar National Commission for Education, Culture and Science. with the continuing support of the partners, the program has led to the establishment of a new Independent center solely for “National Student Capacity Building”. This center will open doors to effective integration of digitalization in STEAM pedagogies, thereby creating opportunities for students to race against the international education standards, ensuring better productivity in terms of research excellence and creative innovation³⁶.
- Knowledge and resource sharing through a series of workshops to create awareness about OER resources and tools available to communities in the MENA region.
- Strengthening the ties with high school students through the National Science Promotion to promote the interests of national youths towards the STEM (Science, Technology, Engineering, and Mathematics) stream. The program shapes the knowledge and collaboration among youth to enrich their research capacity it is opportunity for national high-school students and Qatar University’s undergraduate students to work collectively under the mentorship of highly qualified faculty members and researchers at Qatar University.
- The Qatar University provides course design and development services, consultancy services for efficiently implementing OER. Moreover, to increase awareness on OER the university assigned Digital learning faculty digital learning ambassadors in most Colleges.

Qatar Research, Development and Innovation Council (QRDI)

The QRDI Council, has a range of talent development programs—such as the Graduate Student Research Award (GSRA), Undergraduate Research Experience Program (UREP), High School Research Experience Program (HSREP), and the Displaced Arab Researcher (DAR) program—support the early pipeline of research talent. These initiatives cultivate a culture of scientific inquiry, international collaboration, and research

³⁶ <https://www.wise-qatar.org/project/al-bairaq-qatar/>

integrity, all of which are foundational to open science. Additionally, programs like the Technology Innovation Training Program, Corporate Innovation Leaders, and Government Innovation Leaders help equip emerging innovators with the mindset and skills needed for knowledge sharing, interdisciplinary collaboration, and open innovation. While a formal open science curriculum is still emerging, these initiatives contribute significantly to fostering a new generation of researchers who are prepared to engage with and advance open science values.

4.2. Capacity Building for Policymakers on Integrating Open Science in STI Policies

Qatar has made efforts to build the capacity of policymakers to integrate the core values and principles of open science into national science, technology, and innovation (STI) policies and strategies. While there hasn't been a dedicated national program explicitly focused on open science, there have been significant initiatives aimed at equipping policymakers with the necessary skills and knowledge to support innovation and science policy development. A key example is the Mumakin Program, launched by the QRDI Council, which is designed to build the capacity of targeted policymakers. The program focuses on three core tracks: Intellectual Property (IP) and Technology Transfer, Science and Technology Regulations, and Tools and Methods for RDI Policy Design and Evaluation. Through these tracks, policymakers gain essential training in areas such as IP management, foresight methodologies, and tackling policy challenges. The program also addresses the integration of open science principles into broader policy design, ensuring that stakeholders are equipped to create an inclusive, collaborative, and open research environment. Additionally, through QRDI's foresight training and roundtables on innovation and research policy, there is growing emphasis on aligning national strategies with open science values. These efforts contribute to creating a more open, collaborative, and well-governed innovation ecosystem in Qatar.

4.3. Incorporation of Open Science Competencies in Higher Education Research Curricula

Qatar University (QU) is taking significant steps to prepare researchers for the open science environment by providing them with essential knowledge and skills. The university has incorporated open science competencies into its higher education research skills curricula through a comprehensive approach that includes theoretical frameworks, practical experiences, and technological tools. QU has also launched several initiatives aimed at improving research literacy, fostering innovation, and promoting collaborative scientific practices as follows:

Qatar University Health Sector Curriculum Enhancement: At QU Health, the Academic Quality Department is dedicated to the continuous improvement of both undergraduate and graduate programs. Their focus is on aligning these programs with market needs, accreditation standards, and national priorities³⁷. This approach emphasizes the development of competency-based curricula, ensuring that students gain the essential skills and knowledge needed to excel in their chosen fields.

Development of Research Competencies in STEM by QUYSC: The Qatar University Young Scientists Center (QUYSC) has created the Outcomes-Directed Research Internship Model (ODRIM), based on Research Cognitive Theory (RCT)³⁸. This model combines Course-Based Undergraduate Research Experiences (CURE) with Outcomes-Based Education (OBE) to offer students a thorough research experience. ODRIM has been empirically tested and has shown significant improvements in students' research skills, technical abilities, teamwork, and the dissemination of research outcomes.

Information Literacy Program at Qatar University Library: The QU Library's Information Literacy Program partners with faculty to enhance research skills throughout the campus community³⁹. This program includes instructional sessions for first-year students, core curriculum courses focused on research methodology, and subject-specific library instruction⁴⁰. These initiatives aim to help students effectively identify, evaluate, and utilize scholarly resources, fostering a culture of open and responsible research practices.

³⁷ <https://www.qu.edu.qa/en-us/Colleges/health/Academic-Quality-and-Strategy/Academic-Quality-Department/Pages/curriculum-enhancement.aspx>

³⁸ <https://gcisr.org/qatar-university-young-scientists-center-quysc-researchers-develop-innovative-theoretical-model-for-undergraduate-research-internships/?utm>

³⁹ <https://libguides.qu.edu.qa/InformationLiteracy>

⁴⁰ <https://libguides.qu.edu.qa/c.php?g=281512&p=1875786>

"I am a Researcher" Program by QUYSC: The Qatar University Young Scientists Center (QUYSC) has launched the "I am a Researcher" program⁴¹, which offers national students opportunities for research and participation in international competitions. In partnership with the Ministry of Education and Higher Education (MOEHE), it provides training in research methodology for science teachers in public schools. This program includes hands-on, research-integrated activities designed to enhance research skills and promote skills such as critical thinking, creativity, and innovation among students.

4.4. Use of Open Educational Resources (OER) for Open Science Capacity Building

This section explores the extent to which Open Educational Resources (OER), as defined by the 2019 UNESCO Recommendation on OER, are being strategically leveraged in Qatar for building capacity in open science, and the following outlines the institutions actively integrating OER into their educational and research practices to achieve this alignment.

Qatar University (QU):

The Digital Learning and Online Education (DLOE) Office at Qatar University (QU) held the Second Annual Digital Learning and Online Education Forum, Under the title "The Future of Digital Educational Content in the Era of Artificial Intelligence" in collaboration with UNESCO. It directly supported open science capacity building in Qatar through Open Educational Resources (OER), aligning with UNESCO's 2019 Recommendation. The forum explored OER's potential for enhancing digital learning and research collaboration, emphasizing AI-generated content and open licensing within the framework of open science principles⁴². It equipped faculty and researchers with OER utilization skills to advance open science by promoting open access, open licensing, and knowledge sharing, thereby building institutional capacity for open science in digital learning. The forum is essential for enabling Qatari educators and researchers to adopt OER within the open science ecosystem, ultimately improving the accessibility and dissemination of knowledge in accordance with UNESCO guidelines.

⁴¹ <https://www.qu.edu.qa/en-us/Research/ysc/bridge-program/Pages/am-a-researcher.aspx>

⁴² <https://www.qu.edu.qa/en-us/NewsRoom/Pages/newsdetails.aspx?newsid=13038>

The workshop titled "AI-Generated Educational Content and Open Licensing," organized by the Office for Digital Learning and Online Education at QU, supports the 2019 UNESCO OER Recommendation. It focuses on ethical sharing, Creative Commons licenses, and the integration of Open Educational Resources (OER) with Generative AI. The workshop emphasizes open licensing for AI-generated materials and aims to enhance capacity building in open science by addressing intellectual property and ethical considerations associated with using AI in educational settings. It also highlights the importance of making educational resources accessible while promoting collaborative knowledge sharing within the framework of open science.

The Qatar university took part in the Academic Network for Development Dialogue⁴³, which is an initiative launched by participants from 16 universities and academic institutions, along with representatives from the United Nations Economic and Social Commission for Western Asia (ESCWA), in collaboration with the Academic Council on the United Nations System (ACUNS). The aim of the initiative is to enhance cooperation and knowledge exchange between academic circles and the United Nations in the Arab region. Sessions were offered by Qatar University highlighting OER and scholarly resources that all university students and scholars can use.

Qatar National Library (QNL):

QNL actively supports the creation, use, and dissemination of open educational and research resources. It promotes open access publishing and open licensing, provides institutional repository services (e.g., Manara) to store and share open educational and research content and supports awareness and training around OER best practices.

4.5. Initiatives to Support Open Science Communication

Qatar has implemented several initiatives to foster science communication and disseminate scientific knowledge effectively to researchers across disciplines aligned with open science values and principles. These efforts are outlined below:

⁴³ <https://anddnetwork.org/ar/>

Qatar University:

- **Engagement and Communication Office:** The Engagement and Communication Office at QU provides strategic communications consultancy, events management, and public and media relations planning. It manages the communications and publicity functions of the Vice President for Research & Graduate Studies Office and affiliated centers⁴⁴. Its goals include enhancing the reputation of the Research & Graduate Studies Office by promoting key initiatives, projects, and achievements to various audiences at the local, regional, national, and international levels. They accomplish this through multiple channels, including the Qatar University Research Magazine, newspapers, media outlets, and other publications.
- **Qatar University Research Magazine:** It serves as an essential platform for science communication to disseminate research findings, share knowledge, and promote collaboration on aligning with the values of open science. Its content is accessible to a diverse audience, including researchers, students, and the general public in Qatar. The "Student Projects" section sheds light on research and graduation projects by students from all faculties, encouraging early engagement in scientific communication⁴⁵.
- **QU Press Open Journal System**⁴⁶: This platform hosts Qatar University's academic refereed journals, which provides immediate open access to its content, making all articles freely available and accessible without charge. This open access model facilitates the dissemination of research output from various colleges within the university, making it accessible to a wider research community and the public where users can read, download, copy, distribute, print, search, or link to the full texts of the articles, or use them for any other lawful purpose, without asking prior permission from the journal or the author. In addition, authors retain copyright

⁴⁴ <https://www.qu.edu.qa/en-us/research/offices/engagement-and-communication>

⁴⁵ <https://qu.edu.qa/en-us/Research/publications/Pages/sections.aspx>

⁴⁶ <https://journals.qu.edu.qa/>

to their work, and the publication process is free of charge, promoting the dissemination of scientific knowledge to a broader audience.

- **Qatar University digital hub:** Qatar University Library plays a pivotal role in supporting science communication through its digital hub, which encompasses the 'Qatar University Institutional Repository' and 'Archives and Special Collections'⁴⁷. This centralized platform facilitates the dissemination of Qatar University's original research, scholarly works, and historical materials, thereby promoting the sharing of scientific knowledge across disciplines and to the broader public.
- **Open Education Resources (OER):** QU is dedicated to Open Educational Resources (OER), demonstrating a commitment to providing open access to educational materials, including scientific content⁴⁸.
- **tad Matters® Activities and Events:** it is offered by the office of graduate studies at Qatar University. This initiative includes events like the National 3MT Competition, which challenges graduate students to communicate their research to a general audience within three minutes in an engaging way. This fosters science communication skills among researchers⁴⁹.
- **Research Seminars, Conferences, and workshops:** QU frequently organizes research seminars and conferences, providing opportunities for researchers both within and outside the university to share their findings and engage in meaningful discussions, thereby disseminating scientific knowledge.

Qatar National Library (QNL):

- **Forum for Open Research in MENA (FORM):** Qatar National Library hosted the 3rd annual Forum for Open Research in MENA (FORM) Transforming knowledge into action in October 2024⁵⁰. FORM aims at supporting the advancement of Open Science policies and practices in research communities and institutions

⁴⁷ <https://qspace.qu.edu.qa/?locale-attribute=en>

⁴⁸ <https://www.qu.edu.qa/en-us/Offices/dloe/services/Pages/oer.aspx>

⁴⁹ <https://www.qu.edu.qa/en-us/research/graduate-studies/graduate%E2%80%93learning%E2%80%93support/tadmatters/>

⁵⁰ <https://www.qnl.qa/en/about/news/qatar-national-library-hosts-third-annual-forum-open-research-mena>

across the Arab world. It brings together researchers, policymakers, and innovators from the Middle East and North Africa, fostering discussions on open research practices.

- **Open Access Program:** QNL offers services to support global dissemination and unrestricted access to Qatar-based research, including open access agreements with publishers and development of a digital repository⁵¹.
- **Manara – Qatar Research Repository:** Launched by QNL, Manara is an open platform that captures, preserves, and provides access to scholarly research created in Qatar, promoting global visibility and discoverability across all disciplines⁵².

The Qatar Research, Development, and Innovation Council (QRDI):

- QRDI has undertaken several initiatives to support science communication in line with open science values and principles, with the aim of disseminating scientific knowledge to researchers across various disciplines, decision-makers, and the public. One of the key initiatives is the QRDI Council's National Research, Development, and Innovation (RDI) Communities. These communities provide both physical and virtual platforms for collaboration, driving innovation, and offering access to resources and funding opportunities. They empower like-minded individuals within similar disciplines to showcase their skills and initiatives, fostering collaboration, visibility, and a culture of innovation.
- In addition to the RDI Communities, QRDI's Research Outcome Seminar (ROS) play a pivotal role in open science communication. ROS serves as a platform where academics and grant holders present their research findings, share insights with peers, and engage with communities. This initiative promotes interdisciplinary collaboration and ensures that scientific knowledge reaches a wider audience, including policymakers and the general public. By fostering an environment of openness and exchange, both the RDI Communities and ROS contribute

⁵¹ <https://www.qnl.qa/en/open-access-at-qnl/about>

⁵² <https://www.qnl.qa/en/manara-repository/about>

significantly to a more inclusive and transparent scientific ecosystem in Qatar, furthering the objectives of open science.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1. Initiatives for Reviewing Existing Research Assessment and Evaluation Processes

Qatar University (QU) has the potential to align its research assessment and evaluation processes with open science principles and values by building upon its existing strengths in areas like open access dissemination and quality research initiatives:

- QU has a thorough process for assessing learning outcomes. While primarily aimed at enhancing teaching and program quality, this reflects a dedication to continuously reviewing and improving evaluation methods within the academic field.
- The "Research and Graduate Studies" office at QU focuses on initiatives aimed at improving the quality and impact of research. This includes support for publishing in peer-reviewed journals and promoting excellence in research. A greater emphasis on research impact may also involve considering broader dissemination metrics that align with the goals of open science.
- The presence of QSpace and the QU Press Open Journal System reflects a strong commitment to making research outputs openly accessible. These platforms primarily serve to disseminate research, and their success and usage may indirectly affect how the university values and evaluates research. This could potentially lead to increased recognition of open science practices.

5.2. Open Science Recognition in Career Evaluation and progression systems

Open science practices such as sharing, collaborating beyond academia, and engaging with other knowledge systems are implicitly valued within Qatar University's (QU) career evaluation and progression systems. Several aspects of the current evaluation and

promotion guidelines at QU can be interpreted as supportive of these open science principles such as:

- Collaboration and Engagement with Other Researchers:
 - Co-authored publications are a crucial aspect of promotion guidelines in various colleges, which place significant emphasis on research output. Collaboration is a fundamental element in co-authorship.
 - QU promotes collaborations between industry and academia, which could provide indirect support for the evaluation of researchers.
 - For promotion of research-track faculty, serving as a Principal Investigator (PI) on competitive research grants, whether internal or external, is a crucial factor. obtaining these grants often requires collaboration.
- Dissemination Beyond the Scientific Community (Societal Actors and Public):
 - The promotion guidelines for some QU colleges, include "Services to the community/society" as a criterion for evaluation. This could potentially encompass public engagement activities.
 - Qatar University's commitment to community engagement and outreach, as outlined in the QU SDG report 2023-2024⁵³. The faculty participation in such activities always be regarded positively, even if it is not a formal criterion for promotion.
 - Dissemination through QU Platforms like QU institutional repository, QSpace which facilitate broader dissemination, and faculty who actively contribute to these might be seen positively in terms of their commitment to knowledge sharing.

5.3. Initiatives to recognize and reward open science practices

Recognizing and rewarding open science practices is vital for fostering a collaborative, transparent, and impactful global research ecosystem. This section will highlight key

⁵³ <https://www.qu.edu.qa/en-us/offices/international-affairs-office/documents/qu%20sdg%20report.pdf>

initiatives implemented by institutions in Qatar to recognize open science practices, promoting greater access to and engagement with scientific knowledge.

The primary efforts to acknowledge and reward open science practices in Qatar are led by Qatar University which offers Research Excellence Award⁵⁴ and promotes open science principles by providing the necessary infrastructure and evaluating research impact and collaboration. Additionally, the Qatar National Library, which offers the annual Open Initiative Award⁵⁵ and runs a comprehensive Open Access Program. Moreover, the State Incentive and Appreciation Awards⁵⁶, administered by the Ministry of Culture and Sports, honor Qatari nationals for their outstanding contributions in various fields, including science. These awards recognize works that enhance cultural and intellectual life, promote the dissemination of knowledge, and encourage a culture of excellence.

5.4. Unintended Negative Consequences of Open Science Practices

Open science offers many benefits, but it can also lead to unintended negative consequences. It's essential to acknowledge these potential drawbacks in order to address them effectively. In Qatar, we can identify possible challenges and concerns by examining both the general literature on the unintended negative consequences of open science and the specific context of Qatar's research landscape.

Potential Unintended Negative Consequences in Qatar could be:

- Geopolitical tensions, such as the 2017 blockade of Qatar, have disrupted regional research collaborations. These disruptions hinder the free exchange of knowledge and resources, which are essential for open science. Such political challenges can undermine the effectiveness of open science initiatives by restricting access to collaborative opportunities and shared infrastructure.
- While English is commonly used in scientific communication, making open science resources and research outputs accessible to a wider Arabic-speaking audience in Qatar may pose a challenge. This limitation could decrease the reach and impact of open science within the local community.

⁵⁴ <https://qu.edu.qa/en-us/conference/research-forum/award%E2%80%93categories/research-excellence-award>

⁵⁵ <https://www.qnl.qa/en/open-access-initiative-award>

⁵⁶ <https://www.moc.gov.qa/en/2021/05/29/ministry-of-culture-and-sports-announces-the-reception-of-applications-for-candidacy-for-the-state-incentive-and-appreciation-awards>

- Open data sharing can pose risks of exposing sensitive information and violating privacy if datasets, such as survey results, are not properly anonymized. This risk compromises individual privacy and may lead to legal violations, especially when dealing with personal opinions, health data, or political affiliations. Therefore, researchers must prioritize ethical and legal compliance in data sharing to safeguard participants' rights.
- Openly accessible datasets, if poorly managed, risk misuse like political manipulation or commercial exploitation. Clear data governance policies and ethical guidelines are crucial for responsible open data use.
- Early-career researchers in Qatar might struggle with open science adoption as current academic evaluations favor traditional metrics (publication count, impact factor) over transparency and collaboration.
- The push for open science may compel researchers, particularly those early in their careers, to share findings (such as preprints or open data) before undergoing thorough peer review. This could result in the dissemination of unverified information within Qatar's research and public domains.
- In certain fields within Qatar, where international competition is intense, researchers may be reluctant to openly share early findings or data due to concerns about being "scooped" by researchers from other countries that have more established practices in open science.
- If Qatari research institutions don't explicitly value open science in research assessment and career progression, researchers may be disincentivized to prioritize these activities, as traditional publication metrics currently dominate.

It is important to note that these are potential or inferred consequences. However, being aware of these possible challenges can help Qatar's research and policy communities proactively address them as they further embrace open science principles. Initiatives such as open access publishing through QU Press, QNL's Open Access Program, and discussions at the Forum for Open Research in MENA (FORM), which was hosted in Qatar in October 2024, demonstrate a continued effort to navigate the opportunities and challenges associated with open science in the region.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1. Initiatives for Open Science Methodological and Procedural Changes

Qatar promotes innovative and participatory research methods that enhance openness and align with the UNESCO Open Science Recommendation in section (vi) 21. These initiatives improve accessibility, transparency, and inclusivity throughout the research cycle, emphasizing collaboration and participation.

1. Qatar National Research Fund (QNRF)

QNRF support a number of programs that focus on:

- promoting open science by encouraging researchers to share data openly and transparently to enhance reproducibility and advance scientific progress.
- Supporting cross-border and interdisciplinary collaborations to facilitate knowledge sharing and inclusivity, ensuring that funded research is publicly accessible from the project's outset.

The Qatar National Research Fund (QNRF) has made significant progress in promoting open science throughout the research process. Specifically, regarding sub-point (a) of reference (vi) 21 from the UNESCO Recommendation on Open Science, QNRF's initiatives encourage openness from the very beginning of the research cycle and ensure that this openness continues across all stages. The goal is to enhance quality and reproducibility, foster community-driven collaboration, and support innovative models such as preprints.

QNRF's initiatives contribute to these principles:

- **Open Data Requirements:** From the initial stages of funding, QNRF encourages researchers to openly share their data management plans and commit to making their data available as soon as possible. By integrating openness into the planning and design phases of research, QNRF ensures that the principles of open science are taken into account from the very beginning of each project.
- **Encouraging Openness Across All Stages of the Research Process:** QNRF requires that research results be published openly, promoting transparency throughout the entire research process. This includes making results, methods, and data available for review and reuse. Additionally, QNRF supports open-source platforms for data and research sharing, encouraging researchers to utilize platforms that allow for continuous updates to datasets, methodologies, and findings. This approach creates a dynamic, living document of the research process.

- **Collaborative Research Programs:** QNRF promotes collaboration within the community by funding large, interdisciplinary projects that involve various stakeholders from academia, industry, and the public. This approach leads to more robust and reproducible results.
- **Encouraging Preprints:** QNRF acknowledges the importance of preprints—early, publicly shared versions of research papers that differ from the final peer-reviewed versions. Researchers are encouraged to share their preprints in open repositories before undergoing formal peer review. This practice promotes quicker dissemination of findings and encourages community feedback, which can enhance the quality and reproducibility of research. Preprints also facilitate the identification and correction of potential issues in research methods and conclusions prior to final publication.
- **Preprints and Early Sharing:** QNRF encourages researchers to upload preprints to publicly accessible repositories like arXiv or bioRxiv. This practice enables faster dissemination of scientific knowledge. Preprints are clearly marked as distinct from peer-reviewed publications, allowing the research community to differentiate between preliminary findings and finalized, reviewed work. This approach accelerates knowledge sharing and provides researchers with early feedback that can enhance the quality of their final published work.

2. Qatar National Vision 2030

Qatar National Vision 2030 encourages open science in line with UNESCO Recommendation (vi) 21. It aims to foster open innovation and knowledge sharing while supporting the infrastructure for open access and collaborative research. The initiative prioritizes sustainable development, innovation, and technological advancement, promoting transparency throughout the scientific process, from research to dissemination. It contributes to openness in science as follows:

Sub-point (a): Promoting Open Science from the Outset of the Research Process

Qatar National Vision 2030 highlights innovation and knowledge creation as crucial drivers of national development. This is in line with sub-point (a) of the UNESCO recommendation, which advocates for promoting openness at every stage of the scientific process. It emphasizes community-driven collaboration and participatory research, supported by platforms and initiatives that encourage open data sharing and collaboration. This includes grant schemes that prioritize open access outputs and transparent research processes. Additionally, Qatar's focus on digital infrastructure facilitates the rapid dissemination of knowledge through preprints and open access journals, as recommended by UNESCO.

Sub-point (b): Promoting Open Peer Review and Publication

Qatar National Vision 2030 promotes a knowledge-driven economy, with open science as a fundamental principle. This includes enhancing research standards and implementing open peer review. Initiatives such as Qatar University's adoption of Open Access and QNRF's focus on open access publishing encourage transparent, reproducible, and credible scientific practices. These efforts align with UNESCO's recommendations on open science.

Sub-point (c): Encouraging the Sharing of Negative Results

Vision 2030 promotes open, transparent, and accountable research, in line with UNESCO's recommendation to publish negative results. Qatar is increasingly recognizing the importance of negative results and is encouraging researchers, particularly at Qatar University, to share all research outcomes. This approach aims to enhance scientific integrity and improve the overall quality of research.

Sub-point (f): Supporting the Preservation of Open Data and Resources

Vision 2030 emphasizes the importance of infrastructure for data preservation and open access repositories. The Qatar National Library (QNL) preserves scientific knowledge through digitization and repositories, including Qatar University's QSpace. Additionally,

the Qatar National Research Fund (QNRF) supports research that promotes data sharing and encourages the use of open-source software for preserving and reusing research outputs.

Sub-points (h and i): Encouraging Sharing and Collaboration

Qatar National Vision 2030 promotes interdisciplinary collaboration at national, regional, and international levels, aligning with UNESCO's recommendations for open data resources and research infrastructure access. Qatar's National Research Strategy encourages the creation of collaborative spaces where researchers and members of society can work together to tackle complex challenges, facilitating open data sharing across various sectors. Investments in infrastructure, such as the Qatar National Library (QNL) and Qatar University's open access platforms (like QSpace), support interoperability and broad data sharing, thereby fostering a collaborative research environment.

Sub-point (j): Promoting Open Innovation and Commercialization

Qatar National Vision 2030 promotes innovation by linking science and industry to facilitate the commercialization of research, in line with UNESCO's recommendation on open innovation. The country encourages public-private partnerships that transform research into marketable innovations. The Qatar Science and Technology Park (QSTP) plays a crucial role in promoting open innovation by connecting academia with industry and supporting the commercialization process. Additionally, investments in entrepreneurship enhance open innovation by connecting scientific discoveries to viable market solutions.

3. The Qatar Research, Development, and Innovation (QRDI) Council

QRDI has implemented several national and institutional initiatives to promote innovative, participatory methodological changes throughout the research cycle, enhancing openness in the scientific process, as follows:

The Qatar Research, Development, and Innovation (QRDI) Council, as the custodian of the Innovation track in Qatar's National Development Strategy 3 (NDS-3), has been instrumental in driving these changes. One significant initiative is the Research Outcome Seminars (ROS), where researchers are encouraged to present their findings periodically, not just at the end of their projects. This approach fosters continuous knowledge sharing, peer feedback, and public engagement, aligning with open science principles.

QRDI also strongly advocates for open-source publishing of research findings, ensuring that data and results are accessible to the broader scientific community. This openness facilitates collaboration, reproducibility, and transparency in research.

Additionally, QRDI encourages researchers to move beyond traditional academic dissemination methods. Researchers are expected to utilize social media and other non-traditional methods for sharing their research outcomes. These platforms help engage a broader audience, including industry professionals, policymakers, and the general public, thus democratizing access to scientific knowledge and fostering a more inclusive research ecosystem.

To support these initiatives, QRDI launched Innolight, a comprehensive national online platform that connects key stakeholders in the Research, Development, and Innovation (RDI) ecosystem. Innolight provides a seamless way to explore and access resources, capabilities, and infrastructure within Qatar's innovation and research landscape. It offers features such as RDI Communities for collaboration and an Insights section with dashboards and data-driven intelligence, equipping the RDI community with valuable information to navigate emerging trends and make informed decisions.

Through these initiatives, QRDI is fostering a more open, participatory, and innovative research environment in Qatar, aligning with the goals of NDS-3 and the broader vision of open science.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1. Promoting and facilitating international scientific collaborations on open science

Qatar acknowledges the vital importance of international collaboration in promoting open science, as highlighted in the 2021 UNESCO Recommendation. This section will detail the primary ways in which Qatar is actively fostering and supporting scientific partnerships in open science with global partners, thereby contributing to a more interconnected and accessible research landscape.

- International Research Collaboration Co-Fund (IRCC)⁵⁷: Qatar University's IRCC promotes international research collaboration by funding joint projects that address complex issues. It facilitates the exchange of researchers and students, fostering impactful partnerships that support open science principles like collaboration and knowledge sharing.
- Qatar-Japan Research Collaboration Program (QJRC): It is Funded by Marubeni, this program specifically brings together researchers from QU and Japanese universities to collaborate on topics of mutual interest⁵⁸. The program encourages the development of high-quality, high-impact outcomes that benefit both parties through shared research endeavors which facilitating international scientific collaborations on open science.
- Emphasis on High-Impact Research: QU's strategic goals involve becoming a globally recognized research institution with significant impact. International collaborations are often essential to achieving this, and the resulting research can benefit from open science practices to enhance its reach and influence worldwide.
- The Forum for Open Research in MENA (FORM): Qatar hosts FORM, a leading platform that brings together international researchers, publishers, funders, and policy-makers to align efforts on open science across the Middle East and globally⁵⁹.

⁵⁷ <https://www.qu.edu.qa/en-us/research/research-resources/grants-and-funding/external-funders/qu-ircc>

⁵⁸ [https://www.qu.edu.qa/en-us/research/offices/research-support/grants-and-funding/external-funders/qatar%e2%80%93japan-research-collaboration-program-\(qjrc\)](https://www.qu.edu.qa/en-us/research/offices/research-support/grants-and-funding/external-funders/qatar%e2%80%93japan-research-collaboration-program-(qjrc))

⁵⁹ <https://www.qnl.qa/en/about/news/qln-champions-efforts-promote-open-access-mena-region>

- Qatar National Library Open Access Mandate: Qatar National Library (QNL) support open access to promote international visibility and collaboration. QNL is also a member of international open science coalitions such as the Global Sustainability Coalition for Open Science Services (SCOSS)⁶⁰.

7.2. Strategy for Cross-Border Open Science Collaboration (Best Practices, Capacity Building, Metrics)

This section explores how Qatar promotes the exchange of best practices, builds capacity, and develops metrics for open science through its involvement in regional and international platforms and collaborations.

Qatar University (QU) is promoting international collaboration among multiple stakeholders in open science through various initiatives that facilitate the sharing of best practices, capacity building, and the creation of open science metrics. It collaborates with the United Nations Economic and Social Commission for Western Asia (ESCWA) and several other universities to tackle sustainable development goals through academic research⁶¹. These collaborations promote the sharing of best practices and enhance capacity building in open science. QU is dedicated to incorporating open science practices into its research evaluation processes. The university encourages researchers to publish their work openly, share data transparently, and collaborate with others. By doing so, they contribute to the development of metrics that accurately reflect the impact of open science.

The QRDI 2030 Strategy actively supports cross-border, multi-stakeholder collaboration aligned with open science principles. This includes initiatives focused on international research partnerships, capacity building, and the exchange of best practices. QRDI also engages in global dialogue to inform the development of metrics and frameworks that support openness and collaboration in science.

⁶⁰ <https://scoss.org/representing-the-middle-east-qatar-national-library-joins-scoss/>

⁶¹ <https://www.unescwa.org/news/escwa-and-qatar-university-sign-memorandum-understanding>

7.3. Involvement in Open Science Funding Discussions

Qatar's active participation in open science initiatives and its support for international collaborations indicate its implicit role in shaping the future of open science funding on a larger scale. This section will examine Qatar's potential position and contributions to these ongoing discussions.

Engagement with UNESCO's Open Science Initiatives:

Qatar has shown commitment to UNESCO's efforts in promoting open science. The Qatar Fund for Development (QFFD) signed a trilateral agreement with UNESCO and the Education Above All Foundation to support the United Nations Transforming Education Summit⁶². While this agreement primarily focuses on education, it underscores Qatar's broader collaboration with UNESCO, which includes open science initiatives.

Participation in the Global Research Council (GRC) MENA Regional Meeting:

In November 2023, the Qatar Research, Development, and Innovation Council (QRDIC) hosted the Global Research Council's Middle East and North Africa (MENA) regional meeting in Doha. This event brought together stakeholders to discuss collaborative research funding strategies and the advancement of open science in the region⁶³.

8. GENERAL CONSIDERATIONS

8.1. External Impediments to the implementation of the 2021 UNESCO Recommendation on Open Science.

Whilst Open Science offers numerous benefits across multiple sectors, there are also potential challenges and disadvantages associated with its implementation. These need to be managed and mitigated, otherwise, they could negatively impact on the open science philosophy.

⁶² <https://www.unesco.org/en/articles/unesco-and-qatar-fund-development-sign-agreement-education-above-all-foundation-support-un>

⁶³

https://globalresearchcouncil.org/fileadmin/meetings/documents/2023_Regional_Meetings/2023_GRC_MENA_Regional_Meeting_Report.pdf

Key advantages:

Implementing the recommendations could create a more inclusive and collaborative global scientific community that benefits society. The following could be key advantages of implementation:

- By promoting open access to scientific publications, data, and research outputs, the Recommendation will make scientific knowledge more accessible to researchers, policymakers, educators, and the public. Therefore, as well as increasing accessibility, by helping to bridge the gap between researchers in high-income and low-income countries, it improves equity by providing equal access to information and opportunities for contribution.
- By underpinning a framework for collaboration, open Science encourages interdisciplinary and international cooperation, including between the Global North and Global South. This can lead to more robust and innovative scientific discoveries. These practices also increase transparency in research processes and outputs, helping to build trust in scientific findings and reducing misinformation.
- It is thought that Open Science will lead to faster innovation by reducing barriers to accessing latest scientific data and findings. It will accelerate the pace of research and innovation, and this will be particularly beneficial in responding to global challenges like climate change and pandemics. Freely available scientific knowledge will also contribute to economic growth by stimulating development of new industries, enhancing productivity, and driving progress.
- Open science initiatives will often involve citizen science projects, thereby increasing public engagement and understanding of scientific processes and outcomes, while it will facilitate capacity building by providing opportunities for training and skills development, particularly in regions where access to scientific resources has been limited, e.g. Global South.
- Importantly, it is widely considered that access to open scientific data can inform evidence-based policymaking, leading to more effective and sustainable policies,

especially when improved access to environmental data will support conservation efforts, improve resource management, and monitoring of ecological change.

Key challenges:

While UNESCO's 2021 Recommendation on Open Science offers benefits such as the above, there are also potential challenges and disadvantages. The following are potential key challenges associated with its implementation:

- Open science initiatives often require significant resources, including funding, infrastructure, and human capital, which can be burdensome for under-resourced institutions and countries. The transition to open-access publishing is costly, as the burden moves from readers to the authors who pay publication fees. Fees can be funded through projects, e.g. QRDI/NSF, etc., but they often fall back on the institution.
- Significantly, open access to research findings potentially poses risks related to intellectual property rights, leading to disputes or discouraging innovation due to fears of intellectual property theft. With KTPs, there was an initial reluctance for companies to share their work due to owned IP, and solving this problem led to 'non-disclosure clauses', which effectively defeat the open science aim.
- With increased openness, there is a risk of disseminating low-quality or unverified research, which could undermine trust in scientific outputs. Lack of quality control means some researchers and/or institutions may resist changing traditional practices fearing reputational risks associated with sharing preliminary or inconclusive results.
- While open science aims to reduce inequalities, disparities in digital infrastructure across regions can limit access and participation for some communities. Therefore, solutions are needed for the key pillars of infrastructure and communication.
- Open datasets can potentially be misused if proper guidelines and ethical standards are not followed, leading to incorrect interpretations or harmful

applications. Furthermore, ensuring privacy and security of data, especially personal or sensitive information, can be a major challenge with open science.

- Ensuring the long-term sustainability of open science initiatives requires ongoing commitment and investment, which can be challenging to maintain. Added to this there is an administrative burden for developing and enforcing open science policies and practices which may detract from research time and productivity. All come at a cost!

Wider Context

There are many initiatives which will benefit from implementation of UNESCO's 2021 Recommendation on Open Science, including leveraging science, technology, and innovation (STI) advancements for short and long-term priority issues on the international forefront. Two UN offices – UNESCO International Institute of Higher Education in Latin America and the Caribbean (UNESCO IESALC) and the United Nations Office of South-South Cooperation (UNOSSC) are furthering the role of STI in SDG impacts. Promoting the development of national higher education systems and institutions in member states by ensuring their sustained relevance, quality, efficiency, and equity and advancing regional and international cooperation is among the priorities of UNESCO IESALC's mission in global higher education, with a focus on the Global South.

The recent UNESCO Chairs and Partners Forum: Transforming Knowledge for Africa's Future, was to find ways to strengthen interdisciplinary and collaborative research capacity in Africa, including through more inclusive and effective South-South and South-South-North research partnerships. In the recent Qatar Economic Forum, Eng Omar Al-Ansari, the Secretary-General of QRDI, stated that Qatar were developing research and innovation links with the Global South, which now numbers 134 states from Central and South America, Africa, Asia and Oceania, i.e. the world's developing countries and least developed countries. It is generally considered that the lack of access to current research and scientific knowledge has negatively affected the Global South's innovation ability and economic development.

Further Challenges

Disparities in digital infrastructure across regions provides a further challenge, illustrated by UNICEF's proposed Global Centre of Digital Education. With MoFA and stakeholders on board, lack of clarity/assurance on long term funding has resulted in the project being delayed. Open access publications have placed a burden on educational institutions with professors expecting universities to fund their publications, often without initial discussions. A scheme to publish research by Qatar based academics by Qatar National Library ran out of money within the year, leaving academics to source alternative funding for publication. The demand and cost are very high and maybe unsustainable in the long term. Added to this, there is the potential for researchers to exploit the system and not behave ethically, for example, letting a Qatar based academic submit a journal paper as part of a team on research in another country and getting QNL to fund the publication.

The advantages of implementing UNESCO's 2021 Recommendation on Open Science are clear. However, as well as security implications of sharing cutting edge research, the 'who pays' question will be the critical one which needs to be answered. Addressing challenges will require careful planning, robust policy frameworks, and continued dialogue among stakeholders to maximize benefits while mitigating potential downsides. Provision of the key pillars: scientific knowledge, infrastructure and communication, will also require the 'who pays' question to be answered.